

MIDSEM SYLLABUS

EVERYTHING UPTO & INCLUDING
WHATEVER WILL BE TAUGHT TILL
~~TOMORROW~~ TOMORROW (18/9/2026).

Sizes of sets. $\rightarrow$ sizes of infinite sets.

$\sim$: When two sets ^(A & B) have the same cardinality
then we say that $A \sim B$ or $B \sim A$ and
there is a bijection from $f: A \rightarrow B$ or
 $g: B \rightarrow A$.

$$A \sim B \equiv |A| = |B|$$

Defⁿ Any set A is a finite set if
 $A = \emptyset \rightarrow |A| = 0$.

or
 $A \sim \{1, 2, 3, \dots, n\}$ for some $n \in \mathbb{Z}^+$

$$\hookrightarrow |A| = n.$$

When A is not finite we call it an infinite set.

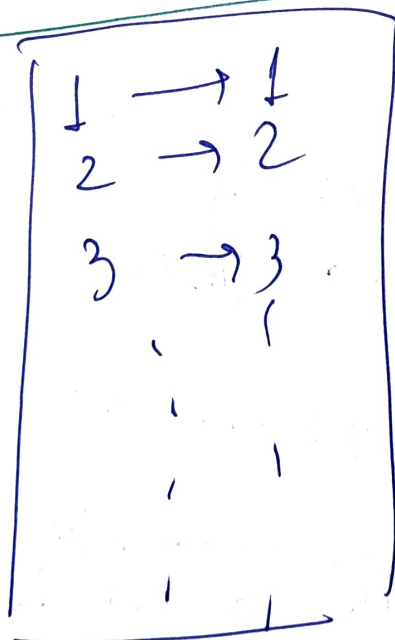
Defⁿ Countable sets.

→ A set S is called countable if $|S| \leq |\mathbb{Z}^+|$

→ There exists an injective 1-to-1 mapping from S to $\mathbb{Z}^+$

→ S is embedded in $\mathbb{Z}^+$

→ A set S is said to be countable if you can make a list/index of its members.
 $\mathbb{Z}^+$ can be itself indexed by itself



Defⁿ

Sequences.

(LCS: Longest common subsequence).

For $n \in \mathbb{Z}^+$ a finite sequence of n terms
is a function whose domain is $\{1, 2, \dots, n\}$,
Such sequences are usually ordered
 $\{x_1, x_2, \dots, x_n\}$
 $f(i) = x_i, 1 \leq i \leq n.$

A infinite $\mathbb{Z}^+$ sequence is a function g having
 $\mathbb{Z}^+$ as its domain.

Such a set is written as

$$\{x_i\}_{i \in \mathbb{Z}^+}$$

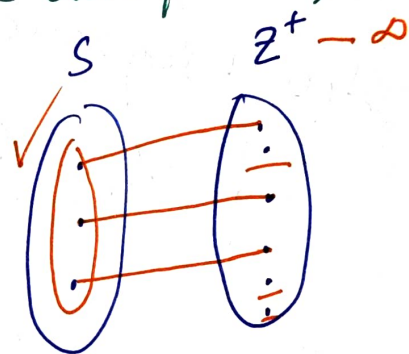
$$\& g(i) = x_i \text{ for } \forall i \in \mathbb{Z}^+$$

Defⁿ

Countably infinite sets.

① A set S is countably infinite if $S \sim \mathbb{Z}^+$

② There exists bijection mapping
from $f: S \rightarrow \mathbb{Z}^+$ (one-to-one correspondence).



E.g.

$$\mathbb{Z}^+ \sim \mathbb{Z}^+$$

$$2\mathbb{Z}^+ \sim \mathbb{Z}^+$$

$$3\mathbb{Z}^+ \sim \mathbb{Z}^+$$

$\therefore 2\mathbb{Z}^+, 3\mathbb{Z}^+$ are all countable.

E.g. $\{1, 1/2, 1/4, 1/8, 1/16\}$.

$f: A \rightarrow \mathbb{Q}^+$ with $A = \{1, 2, 3, 4, 5\}$

$$f(n) = \underline{\underline{2^{-n+1}}}$$

Theorem: If A is a non-empty countable set, then A can be written as a sequence of distinct elements.

Case 1: A is finite

$A \sim \{1, 2, \dots, n\}$ or $\{1, 2, 3, \dots, n\} \sim A$ for $n \in \mathbb{Z}^+$. Define $a_i = f(i)$ for each $1 \leq i \leq n$.

f is bijective $\rightarrow f$ is one-to-one onto $\rightarrow \begin{matrix} \{1, 2, 3, \dots, n\} \\ \downarrow \downarrow \downarrow \\ \{a_1, a_2, a_3, \dots, a_n\} \end{matrix}$

Case 2: A is infinite.

We have a bijective mapping

$$\boxed{g: A \rightarrow \mathbb{Z}^+, g(i) = a_i}$$

$\hookrightarrow$ one-to-one onto

$$\begin{matrix} \{1, 2, 3, \dots\} \\ \downarrow \downarrow \downarrow \\ \{a_1, a_2, a_3, \dots\} \end{matrix}$$

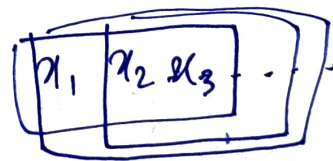


Defⁿ Subsequence:

① The infinite sequence $\{a_1, a_2, \dots\} = \{a_i\}_{i \in \mathbb{Z}^+}$ is a subsequence of $\mathbb{Z}^+$ if for $\forall i \in \mathbb{Z}^+$
 $a_i \in \mathbb{Z}^+$ & $a_i < a_{i+1}$

② Let $\{x_n\}_{n \in \mathbb{Z}^+}$ and $\{y_n\}_{n \in \mathbb{Z}^+}$ be two infinite sequences. We say that $\{y_n\}_{n \in \mathbb{Z}^+}$ is a subsequence of $\{x_n\}_{n \in \mathbb{Z}^+}$ if there exists ~~for~~ a subsequence $\{a_k\}_{k \in \mathbb{Z}^+}$ of $\mathbb{Z}^+$ where for all $k \in \mathbb{Z}^+$ we have

$$y_k = x_{a_k}$$



E.g.

$\{1, 3, 5, 7, \dots\}$ is a subsequence of $\mathbb{Z}^+$

$\{\underline{1}, \underline{2}, \underline{4}, 7, 11, 16, \dots\}$ is a subsequence of $\mathbb{Z}^+$

$$f: \mathbb{Z}^+ \rightarrow \mathbb{Z}^+ \quad a_n = f(n) = 2^{n-1}$$

?

$$\hookrightarrow c_1 = h(1) = 1.$$

$$c_{n+1} = h(n+1) = h(n) + n = c_n + n \text{ for } n \geq 1.$$

$$c_1 = 1, \quad c_2 = c_1 + 1 = 2$$

$$c_3 = c_2 + 2 = 2 + 2 = 4$$

E.g. $\{x_n\}_{n \in \mathbb{Z}^+}$ & $\{y_n\}_{n \in \mathbb{Z}^+}$ be two sequences with
 $n \in \mathbb{Z}^+$ s.t. $x_n = f(n) = (-1)^n + \left(\frac{1}{n}\right)$

$$y_n = g(n) = 1 + \left(\frac{1}{2n}\right)$$

$$\{x_n\}_{n \in \mathbb{Z}^+} = \left\{0, \frac{3}{2}, -\frac{2}{3}, \frac{5}{4}, -\frac{4}{5}, \frac{7}{6}, \dots\right\}$$

$$\{y_n\}_{n \in \mathbb{Z}^+} = \left\{\frac{3}{2}, \frac{5}{4}, \frac{7}{6}, \dots\right\}$$

$$\hookrightarrow y_k = x_{a_k}$$

in this eq.

$$k = n.$$

$$a_k = 2n.$$

$$y_n = \underline{\underline{x_{2n}}}$$

notion of position